

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 3

Duration: 1 Hour

Total Marks:30

Annexure A

Dry Run Evaluation

Code of Conduct:

I Navin Kumar ¹⁹ C192511015) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Navin Kumar . 19

88

88

14.

Constraint Satisfaction problem

Four students A, B, C and D are assigned the time slots 9AM, 10AM and 11AM

Constraints:-

- $A \neq 9AM$
- B must be before C
- $D \neq A$
- $C \neq 11AM$

Variable : A, B, C, D

Domain : $\{9AM, 10AM, 11AM\}$

By ~~constraint~~ propagation

- C cannot be 11AM, so choose $C = 10AM$
- since B must be before C, $B = 9AM$
- A cannot be 9AM, so choose $A = 10AM$
- D must be different from A, so $D = 11AM$

student	Assigned slot
A	10AM
B	9AM
C	10AM
D	11AM

All constraints are satisfied. Hence this is valid CSP solution.

15.

Map colouring CSP

Region A, B, C and D form a cycle

Constraints

- $A \neq B$
- $A \neq C$
- $B \neq D$
- $C \neq D$

Colour: Red, Green and Blue.

Using backtracking

• Assign $A = \text{Red}$

• B is adjacent to A, so assign $B = \text{Green}$

8. c is adjacent to A, so assign $c = \text{Green}$.

A. D is adjacent to both B and c, so assign $D = \text{Red}$.

Region	Assigned Colour.
A	Red
B	Green
C	Green
D	Red

Verification: All adjacent regions have different colours. Therefore the solution is valid.

16.

Cryptarithmic CSP

Solve:-

$$\text{SEND} + \text{MORE} = \text{MONEY}$$

Each letter represents a unique digit from

0-9 and $M \neq 0$

By constraint propagation and backtracking

the valid assignment is

Letter	S	E	N	D	M	O	R	Y
Digit	9	5	6	7	1	0	8	2

Therefore:

$$\begin{array}{r}
 9567 \\
 + 1085 \\
 \hline
 10652
 \end{array}$$

SEND: 9567

MORE: 1085

MONEY: 10652

Verification:

$$9567 + 1085 = 10652$$

Thus the cryptarithmic Csp has a valid solution.

RUBRICS FOR CALCULATION

Numerical Problems						
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	20
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	19
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	21
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	17
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	11

88

88
10